MATE 6551: Tarea 4

Due on December 17, 2025

Prof. Iván Cardona , C41, December 17, 2025

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Problem 0

Compute $H_n(S^0)$ for all $n \geq 0$.

Proof:

Recall that the homology of a space is isomorphic to the direct sum of the homologies of the path components of that space. Therefore, $H_n(S^0) \cong H_n(\{-1\}) \bigoplus H_n(\{1\})$. But, by the dimension axiom, $H_n(\{-1\}) = H_n(\{1\}) = 0$ for $n \geq 1$. We know that the 0th homology of a space is a free abelian group with rank equal to the amount of path components of the space.

Putting it all together, we have:

$$H_n(S^0) = \begin{cases} 0 & \text{if } n \geq 1 \\ \mathbb{Z} \times \mathbb{Z} & \text{if } n = 0 \end{cases} \tag{1}$$

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Problem 1

Prove that P_n is "natural": if $f : X \to Y$ is continuous, there is a commutative diagram

$$\begin{tikzcd} S_n(X) \arrow[r, "P_n^X"] \arrow[d, "f_\#"] & S_{n+1}(X \times I) \arrow[d, "(f \times 1)_\#"] \\ S_n(Y) \arrow[r, "P_n^Y"] & S_{n+1}(Y \times I) \end{tikzcd}$$

Proof:

To show commutativity is to show that $(f \times 1)_\# \circ P_n^X = P_n^Y \circ f_\#$, and to show this, it suffices to show that the equation holds for a generator of $S_n(X)$. Let $\sigma : \Delta^n \to X$ be a singular n-simplex on X, then:

$$\begin{aligned} \big((f \times 1)_\# \circ P_n^X\big)(\sigma) &= (f \times 1)_\#(P_n^X(\sigma)) \\ &= (f \times 1)_\#\big((\sigma \times 1)_\#(\beta_{n+1})\big) \\ &= (f \times 1) \circ \big((\sigma \times 1) \circ \beta_{n+1}\big) \\ &= ((f \times 1) \circ (\sigma \times 1)) \circ \beta_{n+1} \\ &= ((f \circ \sigma) \times 1) \circ \beta_{n+1} \\ &= ((f \circ \sigma) \times 1)_\#(\beta_{n+1}) \\ &= P_n^Y(f \circ \sigma) \\ &= P_n^Y\big(f_\#(\sigma)\big) = \big(P_n^Y \circ f_\#\big)(\sigma) \end{aligned} \tag{2}$$

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Problem 2

If X is a deformation retract of Y, then $H_n(X) \cong H_n(Y)$ for all $n \geq 0$.

Proof:

Suppose that X is a deformation retract of Y. Then there exists a retraction $r : Y \to X$, that is, a continuous map with $r \circ i = 1_X$, where $i : X \to Y$ is the inclusion. Moreover, since it is a deformation retraction, we have $i \circ r \simeq 1_Y$. Applying homology to both equations, and taking advantage of the homotopy axiom, we get:

$$r_* \circ i_* = 1_{H_n(X)}, \quad i_* \circ r_* = 1_{H_n(Y)} \tag{3}$$

$\therefore$ both i_* and r_* serve as isomorphisms between $H_n(X)$ and $H_n(Y)$.

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Problem 3

Prove that the Hurewicz map $\varphi : \pi_1 \to H_1$ is "natural". If $h : (X, x_0) \to (Y, y_0)$ is a map of pointed spaces, then the following diagram commutes:

$$\begin{tikzcd}
\pi_1(X, x_0) \arrow[r, "\pi_1(h)"] \arrow[d, "\varphi_{(X,x_0)}"] & \pi_1(Y, y_0) \arrow[d, "\varphi_{(Y,y_0)}"] \\
H_1(X) \arrow[r, "H_1(h)"] & H_1(Y)
\end{tikzcd}$$

Proof:

To show commutativity is to show that $\varphi_{(Y,y_0)} \circ \pi_1(h) = H_1(h) \circ \varphi_{(X,x_0)}$. Given a path homotopy class $[\gamma] \in \pi_1(X, x_0)$, we start with the left hand size and end up at the right hand side:

$$\begin{aligned}
\left(\varphi_{(Y,y_0)} \circ \pi_1(h)\right)([\gamma]) &= \varphi_{(Y,y_0)}(\pi_1(h)([\gamma])) \\
&= \varphi_{(Y,y_0)}([h \circ \gamma]) \\
&= \text{cls } ((h \circ \gamma) \circ \eta) \\
&= \text{cls } (h \circ (\gamma \circ \eta)) \\
&= \text{cls } \left(h_\#(\gamma \circ \eta)\right) \\
&= H_1(h)(\text{cls } (\gamma \circ \eta)) \\
&= H_1(h)\left(\varphi_{(X,x_0)}([\gamma])\right) = \left(H_1(h) \circ \varphi_{(X,x_0)}\right)([\gamma])
\end{aligned} \tag{4}$$

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Problem 4

Let X be a space and let α, β, γ be (not necessarily closed) paths in X such that $\alpha * \beta * \gamma$ is defined and is a closed path. Prove that, in $H_1(X)$,

$$\text{cls } (\alpha * \beta * \gamma) = \text{cls } (\alpha + \beta + \gamma) = \text{cls } \alpha + \text{cls } \beta + \text{cls } \gamma. \tag{5}$$

Proof:

Note that $\alpha * \beta * \gamma$ is a path in X and thus can be identified with a 1-simplex in X, then $\partial(\alpha * \beta * \gamma) = (\alpha * \beta * \gamma)(1) - (\alpha * \beta * \gamma)(0) = 0$ because it is a closed path. Therefore $\alpha * \beta * \gamma \in Z_1(X) \Longrightarrow \text{cls } (\alpha * \beta * \gamma) \in H_1(X)$.

Now α, β, and γ can also be identified with 1-simpleces in X and so can be added in $S_1(X)$. Note that

$$\begin{aligned} \partial(\alpha + \beta + \gamma) &= (\alpha(1) - \alpha(0)) + (\beta(1) - \beta(0)) + (\gamma(1) - \gamma(0)) \\ &= (\alpha(1) - \beta(0)) + (\beta(1) - \gamma(0)) + (\gamma(1) - \alpha(0)) = 0 \end{aligned} \tag{6}$$

because $\alpha * \beta * \gamma$ is defined and a closed path. Therefore $\alpha + \beta + \gamma \in Z_1(X)$ $\Longrightarrow \text{cls } (\alpha + \beta + \gamma) \in H_1(X)$.

However, cls α, cls β, and cls γ are not necessarily defined, because they are not necessarily closed paths and thus do not necessarily correspond to cycles. The author must be asserting that in the case where cls $(\alpha + \beta + \gamma)$ is defined, he will abuse the notation cls α + cls β + cls γ. Therefore, we prove the first equality and ignore the second.

First, we define a singular 2-simplex $\sigma : \Delta^2 \to X$. We define it on $\dot{\Delta}^2$ as follows:

$$\sigma(1-t, t, 0) = \alpha(t), \quad \sigma(0, 1-t, t) = \beta(t), \quad \sigma(1-t, 0, t) = (\alpha * \beta)(t) \tag{7}$$

Now we extend the definition to the interior of Δ^2 by setting it to be constant on the line segments with endpoints

$$a = a(t) = (1-t, t, 0), \quad b = b(t) = \left(\frac{2-t}{2}, 0, \frac{t}{2}\right) \tag{8}$$

and

$$c = c(t) = (0, 1-t, t), \quad d = d(t) = \left(\frac{1-t}{2}, 0, \frac{1+t}{2}\right) \tag{9}$$

Note that this is well defined and continuous. Furthermore, its boundary is

$$\partial\sigma = \sigma\varepsilon_0 - \sigma\varepsilon_1 + \sigma\varepsilon_2 = \beta - \alpha * \beta + \alpha \tag{10}$$

Now we define another singular 2-simplex $\tau : \Delta^2 \to X$ similarly. We define it on $\dot{\Delta}^2$ as follows:

$$\tau(1-t, t, 0) = \alpha * \beta, \quad \tau(0, 1-t, t) = \gamma, \quad \tau(1-t, 0, t) = (\alpha * \beta) * \gamma \tag{11}$$

Then we extend the definition to the interior by setting it to be constant on the line segments with endpoints $a = a(t), b = b(t)$ and $c = c(t), d = d(t)$. For the same reason, it is well defined and continuous. Its boundary is:

$$\partial\tau = \tau\varepsilon_0 - \tau\varepsilon_1 + \tau\varepsilon_2 = \gamma - (\alpha * \beta) * \gamma + \alpha * \beta \tag{12}$$

Now adding equations (10) and (12) yields:

$$\begin{aligned} \partial(\sigma + \tau) &= (\beta - \alpha * \beta + \alpha) + (\gamma - (\alpha * \beta) * \gamma + \alpha * \beta) \\ &= (\alpha + \beta + \gamma) - (\alpha * \beta) * \gamma \end{aligned} \tag{13}$$

Therefore $(\alpha + \beta + \gamma)$ and $(\alpha * \beta) * \gamma$ differ by a boundary.

$\therefore$ cls $(\alpha + \beta + \gamma) =$ cls $(\alpha * \beta * \gamma)$

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